

CERTA: Quantum-Inspired Tensor-Network Compression with Certified Control for Embedded Vision-Language-Action Models

Shrink a VLA policy with the mathematics of quantum many-body physics, and use the error budget that mathematics provides to put a verifiable safety certificate around what the smaller model does.

Challenge: Volkswagen Group — Quantum-Enhanced VLA Models in AD & Robotics

Team: Entangled Minds

Sub-tracks: Robotics · Compression (primary) + Safety Guarantees (integrated); AD · Compression (cross-track transfer)

Paradigm: Quantum-inspired tensor networks on classical GPUs, hybrid with classical control; simulation-only quantum hardware pathway

Summary. The accepted baseline for VLA compression, INT8 quantisation, halves memory but leaves the parameter count unchanged and gives no tunable accuracy–size trade-off. Pruning and distillation shrink the model but give no principled error budget, and the safety case under ISO 10218 / IEC 62061 still depends on heuristic clipping. We propose **CERTA** (Compression with Error-bounded Rank Truncation and Assurance), which targets two of the challenge's four bottlenecks with one mechanism. **(A)** We decompose the weight matrices of π_0 [7] into **Matrix Product Operators (MPOs)**, the tensor networks behind DMRG in quantum physics. Bond dimensions are allocated from each layer's measured *entanglement spectrum*, and each layer's truncation error is *known exactly* from the singular values we discard. Because MLP hidden neurons can be reordered without changing what the network computes, we reorder them to reduce entanglement before decomposing. This is the quantum-chemistry orbital-ordering trick, and it can help an MPO but can never help an ordinary SVD. **(B)** The compressed policy's action error, calibrated with split-conformal statistics, becomes the bounded disturbance in a **robust Lyapunov / control-barrier-function safety filter**. That turns the accuracy lost to compression into a *budgeted input to a certificate* instead of an unquantified risk. Targets: $\geq 2\times$ fewer parameters (stretch $2.5\times$) at $\leq 5\%$ relative LIBERO success drop, $\geq 4\times$ smaller memory footprint with INT8 stacked, $\geq 95\%$ of perturbed episodes ending in a certified safe equilibrium, and a mandatory ablation against SVD and pruning at matched parameter count, reported whether it favours us or not.

1. Problem Framing

Our reading of the problem. The challenge statement names the core difficulty as the *compound constraint*: a deployed VLA must be small enough for a Jetson-class controller, fast enough for a 100 ms cycle, *and* able to support a safety argument, all at once. Existing methods address one of these at a time. INT8 halves memory, but the 3.3B-parameter π_0 still has 3.3B parameters and still needs an 8-bit datapath. Distillation and pruning change size, but the accuracy they lose is found empirically after the fact, with no error budget beforehand. On the control side, the accepted safety baseline, clipping at joint and velocity limits, is a heuristic that stops a violation already under way; it does not prove the arm will settle in a safe state. The result is a model that is either too big to deploy or too uncertain to certify. We treat “how much accuracy did compression cost?” and “how much disturbance must the safety layer tolerate?” as **the same quantity**, measured once and used twice.

Why we lead with robotics, and why AD is included. Among the sub-tracks, Robotics · Compression has the best-specified reference setup (π_0 , public base and LIBERO-fine-tuned checkpoints, INT8 baseline) and a *closed-loop* task metric, success rate in simulation, rather than an offline proxy. That lets a 3–4 month sprint produce evidence about what the policy actually does rather than a perplexity-style number. Volkswagen Group runs large fleets of industrial and collaborative robots, so this is a direct use case, not a stand-in. The same pipeline is designed to be model-agnostic; we commit to repeating it on LLaVA-1.5-7B for the AD compression sub-track (§6), whose Llama-2-7B-family decoder is the exact architecture on which tensor-network compression evidence already exists [1].

Why quantum-inspired tensor networks are a credible advantage, not a label. Three reasons, each testable early in the PoC:

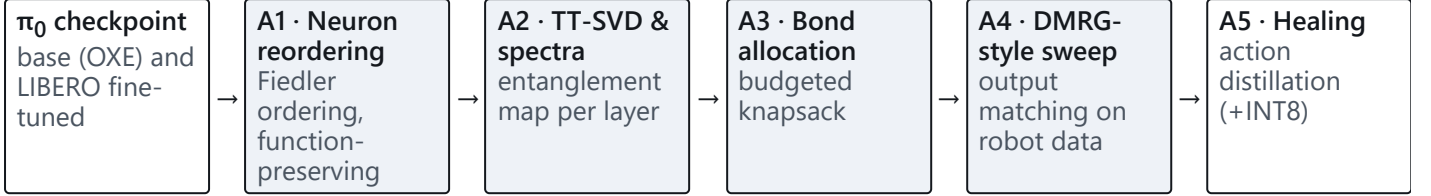
- **Multi-scale structure SVD cannot see.** A 2048→16384 Gemma MLP projection is an operator from 2^{11} to 2^{14} dimensions. Its indices factor naturally into “qubit-like” legs. A rank- r SVD captures correlation across *one* bipartition; an MPO captures a sequence of bipartitions. Wherever singular spectra decay across several cuts (the analogue of the physics *area law*), an MPO stores more of the operator per parameter. CompactifAI reported this on LLaMA-2-7B: about 70% fewer parameters with a 2–3% accuracy drop [1].
- **A gauge freedom that plain SVD cannot use.** Reordering an MLP's hidden neurons leaves the network's function unchanged, and SVD's result is identical under any reordering. An MPO's bond entanglement *does* depend on ordering. DMRG in quantum chemistry cuts entanglement by ordering orbitals along the Fiedler vector of a mutual-information graph [11]. We do the same with neurons. If this helps, the gain is attributable to tensor-network structure specifically, and our ablation isolates it.
- **Errors that are known, not estimated.** TT-SVD truncation obeys $\|W - W_{\text{MPO}}\|_F \leq (\sum_k \delta_k^2)^{1/2}$, where δ_k is the discarded singular weight at cut k [8]. Every step in bond dimension has a known cost, which makes compression a

budgeted optimisation instead of a trial-and-error sweep, and gives the safety layer (§2B) something principled to calibrate against.

What we do not claim. No asymptotic quantum speed-up, and no dependence on fault-tolerant algorithms such as [2]. Tensor networks are *quantum-inspired* and run on GPUs. The claim we will defend: at a matched parameter count, entanglement-guided MPO compression keeps more closed-loop task success than activation-aware SVD and structured pruning, and its error budget supports a safety certificate that heuristic clipping cannot. If the ablation shows no MPO advantage, we will report that as the headline result.

2. Technical Approach

OFFLINE — compression and calibration (GPU; no latency constraint)



ONLINE — per control cycle (≤ 100 ms budget)

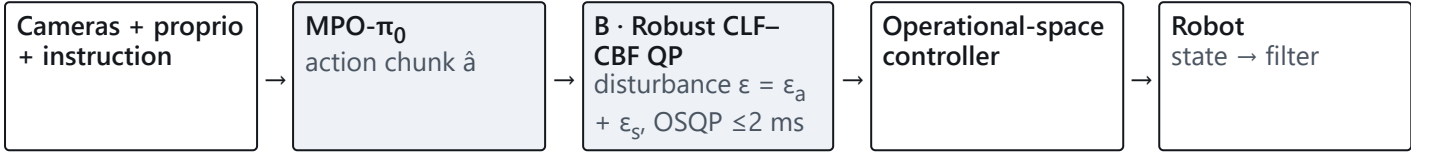


Figure 1. CERTA pipeline. Shaded blocks are tensor-network-derived. The conformal action-error bound ϵ_a is measured offline on the compressed model and consumed online by the safety filter. That link is what makes this a single integrated method rather than two separate projects.

A. Entanglement-guided MPO compression (primary)

A1 — Entanglement-reducing neuron ordering. For each MLP block we build a graph over hidden neurons weighted by activation mutual information on calibration data (LIBERO/OXE observations, not generic text). We then permute the neurons by the graph's Fiedler vector, applying the same permutation to the gate, up and down projections. The network's outputs are bit-identical after this step; only the tensor-network cost changes.

A2 — MPO factorisation. Each weight $W \in \mathbb{R}^{m \times n}$ is reshaped over factorised index legs (for Gemma: $2048 = 4 \cdot 4 \cdot 4 \cdot 4 \cdot 8$, $16384 = 8 \cdot 8 \cdot 8 \cdot 8 \cdot 4$) and decomposed by TT-SVD into cores with $\sum_k \chi_{k-1} m_k n_k \chi_k$ parameters [8, 9]. At each cut we record the singular spectrum and the von Neumann entropy $S = -\sum p_i \log p_i$, which gives a compressibility map across SigLIP, Gemma and the action expert. The vocabulary embedding (≈ 527 M parameters, mostly unused by a robot policy) is compressed as a TT-embedding [10].

A3 — Budgeted bond-dimension allocation. We choose χ_ℓ for every layer to minimise $\sum_\ell s_\ell \cdot \epsilon_\ell(\chi_\ell)^2$ subject to a global parameter budget. Here ϵ_ℓ is the exact truncation error from A2 and s_ℓ is a Fisher-style sensitivity of the *flow-matching action loss*. This is a multiple-choice knapsack, solved exactly by dynamic programming. We use a classical solver on purpose, because a quantum optimiser would add no value at this size. The action expert and final projections are protected by default.

A4 — Calibration-aware DMRG-style sweep. Weight-space truncation ignores which input directions actually occur. Following the alternating (DMRG/ALS) scheme [12], we sweep across each layer's cores and update them to minimise $\|(W - W_{\text{MPO}})X\|_F$ on calibration activations X , re-truncating at each two-site update so the bond allocation stays within budget. This is the tensor-network counterpart of activation-aware SVD [13], which is also our strongest control.

A5 — Healing and stacking. A short fine-tune of the MPO cores only, using the flow-matching loss plus distillation toward the uncompressed teacher's action chunks, recovers residual error. INT8 quantisation (bitsandbytes) of the cores is then applied on top. The two methods are complementary: MPO cuts parameters, INT8 cuts bytes per parameter.

B. Certified action filter: Lyapunov and control-barrier functions under a compression error budget (integrated)

Error budget. On held-out calibration episodes we compute the split-conformal quantile ϵ_a of $\|\hat{a}_{\text{MPO}}(o) - a_{\text{ref}}(o)\|$, which yields $P(\|\Delta a\| \leq \epsilon_a) \geq 1 - \alpha$ without assumptions on the network [14]. Sensor and tracking error ϵ_s is calibrated the same way. **Filter.** Barrier functions $h_i(x)$ encode joint-limit margins, end-effector speed limits, workspace bounds and keep-out planes. Each cycle a QP solved with OSQP [16] finds the command closest to the policy's action such that $\nabla h_i \cdot (f + gu) \geq -\kappa h_i + \|\nabla h_i g\| \epsilon$ (robust CBF [15]). When a constraint saturates or an out-of-distribution score trips, a Lyapunov constraint $\dot{V} \leq -\gamma V + c\epsilon$ with $V(x) > 0$ drives the arm to a safe resting equilibrium. **What this proves.** For the nominal robot model with disturbance bounded by ϵ , the safe set is forward-invariant and the state converges

to a small neighbourhood of the safe equilibrium (input-to-state stability). The conformal bound holds with probability $1-\alpha$ in distribution; under shift it may not hold, so we *measure* coverage at each perturbation severity rather than assume it. **Certificate-aware compression.** Adding $\epsilon_a \leq \epsilon_{\max}$ as a constraint in A3 produces a *compression-safety Pareto front*: how small the model can get before its error budget shrinks the usable safe set.

C. Quantum hardware pathway (secondary objective, simulation only)

Left-canonical MPS cores are isometries, and a bond dimension $\chi = 2^b$ maps to a sequential quantum circuit with b bond qubits [17, 18]. For representative cores we report qubit count, transpiled depth and two-qubit gate count, and noise sensitivity under Qiskit Aer depolarising models. We expect this to show that gate cost grows as $\sim 4^b$, so near-term hardware is plausible only for $\chi \leq 16$ components. That is a useful, honest boundary, not a promise.

3. Feasibility and Resource Requirements

Parameter budget (planning estimate from the published π_0 architecture). Gemma-2B MLPs dominate the parameter count, which is why the method focuses there:

Component of π_0	Params (M)	Treatment	Target ratio	After (M)
Gemma-2B MLP blocks (18×3 projections)	1,812	MPO, A1–A5	$4\times$	453
Vocabulary embedding ($257k \times 2048$)	527	TT-embedding	$\approx 10\times$	53
SigLIP So400m MLPs	268	MPO	$2\times$	134
Gemma attention, SigLIP attention, action expert, misc.	≈ 690	Protected (bf16)	$1\times$	≈ 690
Total	$\approx 3,300$	—	$\approx 2.5\times$	$\approx 1,330$

Memory: bf16 ≈ 6.6 GB $\rightarrow$ INT8 baseline ≈ 3.3 GB $\rightarrow$ MPO bf16 ≈ 2.7 GB $\rightarrow$ MPO + INT8 ≈ 1.4 GB. The committed target is $2\times$ (≤ 1.65 B parameters); $2.5\times$ is a stretch, because the 2B-scale backbone is likely less redundant than the 7B models in [1].

Resource	Requirement (estimate)	Basis
Training compute	≈ 900 A100/H100-80GB GPU-hours, ≈ 430 kWh (PUE 1.2); $\approx$ US\$2–3k at cloud rates	≈ 33 healing runs $\times$ 16 GPU-h; LIBERO evaluation ≈ 100 GPU-h; AD transfer ≈ 150 GPU-h
Latency profiles	$1\times$ RTX 4090 workstation; $1\times$ Jetson AGX Orin 64GB	Desktop reference (π_0 reports ≈ 73 ms [7]) and embedded target
Software (all open)	openpi / LeRobot (π_0 , PyTorch), LIBERO + robosuite/MuJoCo, bitsandbytes, tntorch/quimb, OSQP, Qiskit Aer, Docker	Recommended toolset in the challenge statement, §6
Data	LIBERO (4 suites), Open X-Embodiment held-out split, nuScenes + NuScenes-QA [19]	Public; used under each licence; no VW data required
Quantum hardware	None. Tensor networks run on GPUs; pathway analysis runs in Qiskit Aer	Per challenge §5.3

Assumptions and constraints. (i) The MPO dimension factorisations are workable with contraction order tuned for GPU throughput. If native contraction is slower than dense layers at our bond dimensions, we fuse adjacent cores with torch.compile/Triton, and all latency and memory figures are reported *in the deployed format*. (ii) π_0 PyTorch checkpoints reproduce published LIBERO success rates to within 3 percentage points; this is Gate G1. (iii) The safety certificate applies to the simulated Franka Panda model; sim-to-real transfer and formal certification are out of scope for Phase 2 and will be stated as such. (iv) Conformal coverage assumes calibration and test data are exchangeable; we measure where that breaks down.

Risk	Likelihood	Mitigation / fallback
MPO is no better than activation-aware SVD at matched parameters	Medium	Gate G2 (week 7) checks this before the full grid. The negative result is published, and the best compressor is still delivered, clearly labelled
2B-scale backbone less compressible than 7B evidence suggests	Medium	Report the Pareto curve rather than a single point; shift budget to the embedding and SigLIP; cross-check on the 7B AD model
Conformal bound fails under severe perturbation	Medium–high	Report coverage by severity; OOD score triggers the conservative Lyapunov safe-stop mode; guarantees stated only for the bounded regime
Healing exceeds compute budget	Low–medium	LoRA on MPO cores; fewer steps with teacher distillation; ablations at a single compression level

4. Expected Impact

What a successful PoC demonstrates. First, that a generalist robot VLA can be made **2–2.5× smaller in parameters and $\geq 4\times$ smaller in memory** while keeping closed-loop task success within the challenge's 5% threshold. That moves a π_0 -class policy from workstation GPUs toward Orin-class robot controllers and cobot compute units. Second, and more distinctive: **the accuracy given up for compression becomes a quantified, conformally calibrated disturbance bound that a Lyapunov/CBF filter is provably robust to.** That gives safety engineers an artefact that fits into an ISO 10218 / IEC 62061 argument, which “we clipped at the joint limit” does not. The method is model-agnostic, so the same pipeline and certificate apply to AD VLAs under ISO 26262. Cross-track transfer is a committed deliverable (§6).

Target (pre-registered)	Success threshold	Measured on / against
Compression vs accuracy	$\geq 2\times$ parameter reduction at $\leq 5\%$ relative success drop (stretch 2.5 $\times$); full Pareto curve at 1.5/2/2.5/3 $\times$	LIBERO 4 suites, 500 episodes each; vs bf16 and INT8
Quantum-inspired attribution	MPO success > best of SVD / pruning at matched parameters, 95% CI excluding zero, on ≥ 3 of 4 suites	Ablation rows 3–7 (§5)
Offline accuracy (challenge metric)	$\leq 5\%$ relative increase in action error at $\geq 2\times$	Open X-Embodiment held-out split, π_0 base
Memory footprint	$\geq 4\times$ below bf16, $\geq 2\times$ below INT8 baseline	Deployed checkpoint size and peak GPU memory
Latency (reference profile)	≤ 100 ms end-to-end including filter, and not slower than bf16	RTX 4090; Orin measured and reported without a pre-committed target
Efficiency gain	$\geq 10\%$ FLOPs reduction per inference; $\geq 25\%$ fewer GPU-hours to adapt to a held-out task	Profiler FLOPs; LIBERO-10 adaptation vs full AdamW + cosine fine-tune
Safety guarantee	$\geq 95\%$ of perturbed episodes end in a safe equilibrium; zero constraint violations inside the certified regime	Synthetic perturbation suite vs heuristic joint/velocity clipping
Cross-track transfer	$\geq 2\times$ parameters at $\leq 5\%$ relative accuracy drop	LLaVA-1.5-7B on NuScenes-QA vs INT8

Value under a negative result. If MPOs do not beat SVD at matched size, the PoC still delivers a compressed π_0 meeting the challenge thresholds with the best available method, the first per-layer entanglement map of a VLA (which shows *where* these models are redundant), a reusable compression-plus-certificate harness, and a clear answer to whether quantum-inspired structure pays off for embodied models.

5. Validation Plan

5.1 Protocol and statistics

- **Closed-loop first.** LIBERO Spatial, Object, Goal and Long: 10 tasks $\times$ 50 fixed initial states per suite, identical across all arms so comparisons are *paired*. OXE offline action error and NuScenes-QA accuracy are secondary.
- **Seeds and reporting.** Every arm runs ≥ 3 independent seeds (calibration sample, healing, simulator), reported as mean $\pm$ SD, with 95% bootstrap CIs over task $\times$ seed. The $\leq 5\%$ drop is tested one-sided for non-inferiority. “Relative” drop is used wherever it is stricter than absolute percentage points.
- **Matched budgets.** Every compressed arm gets the same parameter count per component, the same calibration data and the same number of healing steps. Hyperparameters are tuned on LIBERO-90 tasks disjoint from evaluation. The evaluation set is run once, after decisions are frozen.

5.2 Ablation ladder (mandatory quantum-inspired isolation)

# Arm (all at matched parameter count where compressed)	What it controls for
1 π_0 bf16	Accuracy reference
2 INT8, bitsandbytes	Accepted challenge baseline
3 Activation-aware truncated SVD [13] + same healing	“Any low-rank method works”: key control
4 Structured MLP-width pruning + same healing	“Any shrinkage works”
5 MPO, random neuron order, uniform χ	Tensor-network structure alone
6 MPO + A1 ordering + A3 allocation, no A4 sweep	Contribution of the data-aware sweep
7 CERTA MPO, full A1–A5 (treatment)	—
8 CERTA MPO + INT8	Stacking benefit
9 bf16 π_0 fine-tuned with the same healing steps	That healing is not itself improving the policy

Only the gap between row 7 and the better of rows 3–4 is credited to the quantum-inspired component. Rows 5–6 break that gap down further. Safety has its own ladder: no filter; heuristic clipping (baseline); non-robust CBF ($\epsilon=0$); robust CLF–CBF with ϵ from the *uncompressed* model; robust CLF–CBF with ϵ from the *MPO* model (treatment).

5.3 Safety evaluation

Perturbation suite at three severities: image noise, blur and brightness; camera pose offset ($\pm 2\text{--}5$ cm, $\pm 3\text{--}8^\circ$); proprioceptive noise; action delay of 1–3 steps; object mass $\times 0.5\text{--}2$ and friction $\times 0.5\text{--}1.5$; external end-effector force impulses; paraphrased instructions. **Safe equilibrium** (defined in advance): at episode end, joint-velocity norm < 0.05 rad/s held for ≥ 0.5 s, state inside every barrier set, and no violation during the episode. We report the % of safe-equilibrium episodes, task success under perturbation, filter intervention rate, empirical conformal coverage by severity, and QP solve time.

5.4 Timeline and decision gates (16-week sprint)

Weeks	Work	Exit criterion
1–3	Containerised baselines: π_0 bf16 and INT8 on LIBERO and OXE; latency harness; perturbation suite	G1: LIBERO within 3 pp of published
4–7	Entanglement maps; A1–A5 at $2\times$; SVD and pruning controls at $2\times$, 1 seed	G2: MPO $\geq$ SVD on $\geq 3/4$ suites; otherwise pivot to a hybrid and report
8–11	Full ablation ladder, 3 seeds; Pareto curve; INT8 stacking; 4090 and Orin latency. In parallel: safety filter and conformal calibration	Compression targets met or characterised
12–14	Safety ladder on perturbation suite; compression–safety Pareto; AD transfer (LLaVA-1.5-7B); hardware pathway	Safety and transfer targets met or characterised
15–16	Clean-VM replication by a team member who did not write the code; Apache-2.0 release; 4–8 page report; resource declaration	One-command end-to-end reproduction

6. Hybrid / Cross-Domain Integration

Each technique does what it is good at. Tensor networks provide structure and exact error accounting. Gradient-based healing recovers what truncation cannot. INT8 cuts bytes per parameter. Classical robust control turns a statistical error bound into a guarantee about the robot's behaviour. None of these components is asked to do the others' job, and no quantum processor appears in the 100 ms loop. **HPC/AI integration:** decomposition, sweeps and healing run as standard PyTorch/FSDP jobs; MPO layers are drop-in `nn.Module` replacements, so the compressed policy runs in the unmodified openpi/LeRobot serving stack and exports to TensorRT. **Cross-track transfer:** the pipeline is repeated on LLaVA-1.5-7B for AD compression (NuScenes-QA, INT8 baseline). The filter design carries over to trajectory outputs, with lane, time-to-collision and acceleration barriers in place of joint limits, and we describe that mapping for ISO 26262 in the report. **For VW specifically:** one compression and certification toolchain covering both factory robotics and in-vehicle VLAs.

7. Team Capability

Member	Role on this project	Relevant expertise
Kyoung-Tae (Elijah) Kim MSc Quantum Computing, UPM (Spain)	Quantum Project Manager: sprint plan and decision gates G1/G2; orchestration of the ML pipeline (calibration, healing, evaluation harness, experiment tracking)	AI & ML, RAG, LangChain, agentic-AI orchestration and system optimisation
Naveed Qureshi MSc Quantum Computing, UPM (Spain)	Quantum Technical Lead: MPO decomposition, entanglement analysis, budgeted bond-dimension allocation (A1–A4), quantum hardware pathway (C)	Satellite mission-planning optimisation using QUBO; constrained discrete optimisation
Alejandro Valencia Blancas MSc Quantum Computing, UPM (Spain)	Quantum Solutions Architect: end-to-end software architecture, safety-filter integration (B), latency profiling, reproducible open-source release	Software engineering, JavaScript, database systems, mobile communications, AI; ethics and epistemology of AI

Why this team. Entangled Minds formed on the MSc in Quantum Computing at the Universidad Politécnica de Madrid (UPM), where the three of us worked together on group projects throughout the programme. Across those quantum projects, spanning quantum algorithms, variational and quantum-inspired methods, and quantum optimisation, we developed the working method this proposal depends on: design hybrid quantum–classical systems, validate in simulation first, and benchmark honestly against strong classical baselines. Our roles are complementary and map directly onto CERTA’s workstreams. Naveed has formulated satellite mission planning as a QUBO, a budgeted

allocation problem under hard constraints, which is the same class of problem as bond-dimension allocation (A3) and the hardware-pathway analysis. Elijah’s experience orchestrating and optimising multi-stage AI systems covers the compress–heal–evaluate pipeline and its roughly 33 tracked runs. Alejandro brings the software-engineering depth needed for the challenge’s one-command, clean-environment reproducibility requirement, and his background in the ethics and epistemology of AI grounds the safety-assurance framing (ISO 10218, IEC 62061, EU AI Act).

Execution readiness. Large-scale VLA fine-tuning and robot control lie outside our core experience, so we address them through the design of the plan itself. We build on maintained open-source stacks (openpi/LeRobot, LIBERO, OSQP) rather than custom infrastructure. We stage the sprint so that weeks 1–7 produce stand-alone deliverables (reproduced baselines, the entanglement map, and a go/no-go decision on MPO vs SVD) before any expensive grid runs. And a team member who did not write the code carries out the clean-environment replication. Compute of about 900 GPU-hours (estimated US\$2–3k) will be sourced through cloud GPU rental or academic allocations.

References. [1]–[7] as numbered in the VW challenge statement §7 (CompactifAI; Liu et al. 2024; Alpamayo-R1; NVFP4; Open X-Embodiment; nuScenes; π_0). [8] Oseledets, SIAM J. Sci. Comput. 33 (2011) · [9] Novikov et al., Tensorizing Neural Networks, NeurIPS 2015 · [10] Hrinchuk et al., Tensorized Embedding Layers, Findings EMNLP 2020 · [11] Barcza et al., PRA 83, 012508 (2011) · [12] Holtz, Rohwedder & Schneider, SIAM J. Sci. Comput. 34 (2012) · [13] Wang et al., SVD-LLM, arXiv:2403.07378 · [14] Angelopoulos & Bates, arXiv:2107.07511 · [15] Jankovic, Automatica 96 (2018) · [16] Stellato et al., OSQP, Math. Prog. Comp. (2020) · [17] Schön et al., PRL 95, 110503 (2005) · [18] Huggins et al., Quantum Sci. Technol. 4, 024001 (2019) · [19] Qian et al., NuScenes-QA, AAAI 2024 · Liu et al., LIBERO, NeurIPS 2023.

Appendix — Supplementary Material

CERTA: Quantum-Inspired Tensor-Network Compression with Certified Control for Embedded Vision-Language-Action Models

Challenge: Volkswagen Group — Quantum-Enhanced VLA Models in AD & Robotics

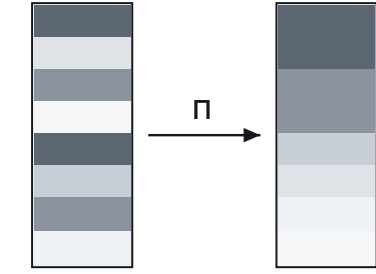
Team: Entangled Minds

Contents: A. Technical diagrams · B. Mathematical specification · C. Validation specifications · D. References

Supplements the 6-page concept proposal; section references (§) point to the proposal. All material comes from public literature, open-source software and the team's own design. No confidential or proprietary information is included.

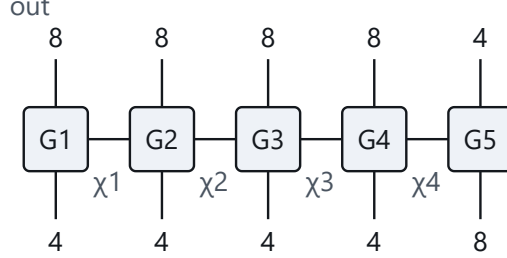
A. Technical Diagrams

(a) Reorder hidden neurons



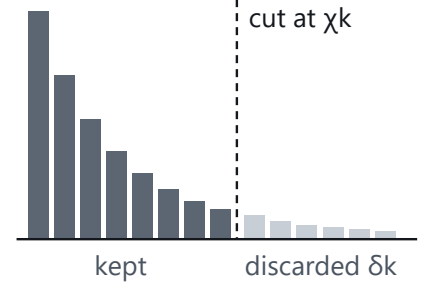
$W_{up} : 16384 \times 2048$
gate/up rows $\leftarrow \Pi \cdot W$
down columns $\leftarrow W \cdot \Pi^{-1}$
output bit-identical

(b) Factorise as an MPO



in
out
out legs $8 \cdot 8 \cdot 8 \cdot 8 \cdot 4 = 16384$
in legs $4 \cdot 4 \cdot 4 \cdot 4 \cdot 8 = 2048$
bonds: $\chi_1, \chi_4 \leq 32$; $\chi_2, \chi_3 \leq 1024$

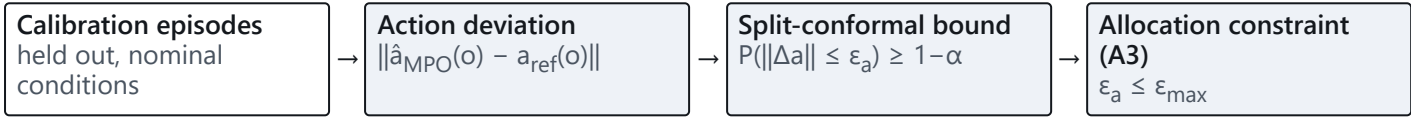
(c) Truncate every cut



$\|W - W_{MPO}\| \leq \sqrt{\sum_k \delta_k^2}$
entropy $S = -\sum p \ln p$
 $p = \sigma^2 / \sum \sigma^2$

Figure A1. MPO compression of one Gemma-2B MLP projection (steps A1–A3). **(a)** Applying the same permutation Π to the gate, up and down projections leaves the network's output unchanged. **(b)** The 16384×2048 matrix becomes five cores; the bond dimensions χ_k set the size–accuracy trade-off. **(c)** At every cut, the discarded singular weight δ_k is known exactly and bounds the total error. An ordinary SVD has a single cut and its spectrum does not change under Π . Any gain from reordering is therefore attributable to the tensor-network structure.

OFFLINE — calibrate the error budget



ONLINE — enforce every cycle, then measure



Figure A2. From compression error to safety certificate (§2B). The guarantees hold for the nominal robot model with disturbance bounded by $\varepsilon = \varepsilon_a + \varepsilon_s$. Whether that bound actually holds is measured at each perturbation severity, not assumed.

B. Mathematical Specification

B1. MPO form, parameter count and error bound

Split the row and column indices of $W \in \mathbb{R}^{m \times n}$ into legs $m = \prod m_k, n = \prod n_k$. An MPO writes each entry as a product of matrices: $W(i_1 \dots i_N; j_1 \dots j_N) = G_1[i_1, j_1] G_2[i_2, j_2] \dots G_N[i_N, j_N]$, with $G_k[i, j] \in \mathbb{R}^{\chi_{k-1} \times \chi_k}$ and $\chi_0 = \chi_N = 1$. The parameter count is $P = \sum_k \chi_{k-1} m_k n_k \chi_k$, and no bond can exceed the smaller side of its cut: $\chi_k \leq \min(\prod_{1 \leq l \leq k} m_l n_l, \prod_{1 \leq l \leq k} m_l n_l)$. If TT-SVD discards singular weight δ_k at each of the $N-1$ cuts, then $\|W - W_{MPO}\|_F \leq (\sum_k \delta_k^2)^{1/2}$ [8]. The entanglement entropy at cut k is $S_k = -\sum_i p_i \ln p_i$, where $p_i = \sigma_i^2 / \sum_j \sigma_j^2$.

B2. Why reordering preserves the function but changes the MPO

A Gemma MLP computes $y = w_{down}[\phi(w_{gate}x) \odot w_{up}x]$. For any permutation matrix Π , replace $(w_{gate}, w_{up}, w_{down})$ with $(\Pi w_{gate}, \Pi w_{up}, w_{down} \Pi^{-1})$. Because ϕ and $\odot$ act element by element, $\phi(\Pi a) \odot \Pi b = \Pi[\phi(a) \odot b]$, so y is unchanged. Π is orthogonal, so the singular values of ΠW equal those of W and SVD compression is unaffected. MPO bond spectra, however, depend on which neuron is assigned to which leg position. We order neurons by the Fiedler vector of the graph Laplacian $L = D - A$, where $A_{ij} = -\frac{1}{2} \ln(1 - \rho_{ij}^2)$ is the Gaussian mutual information between

neurons i and j computed from activation correlations on robot calibration data. This follows the orbital-ordering heuristic from quantum chemistry [11].

B3. Budgeted bond-dimension allocation and the robust safety filter

Allocation. Minimise $\sum_{\ell} s_{\ell} \epsilon_{\ell}(\chi_{\ell})^2$ subject to $\sum_{\ell} P_{\ell}(\chi_{\ell}) \leq B$, choosing each χ_{ℓ} from a set of 16 candidate values. Here s_{ℓ} is the empirical Fisher sensitivity of layer ℓ , the expected squared gradient norm of the flow-matching loss with respect to w_{ℓ} . This is a multiple-choice knapsack problem. With the budget discretised in steps of 10^5 parameters, dynamic programming over the ≈ 109 compressible matrices takes about 2×10^7 operations, so it is solved exactly on a CPU. For the safety-constrained variant, we add $\epsilon_a \leq \epsilon_{\max}$ through a Lagrangian sweep. **Filter.** The robot is modelled as control-affine, $\dot{x} = f(x) + g(x)(u + d)$ with $\|d\| \leq \epsilon$. Each cycle solves $u^* = \operatorname{argmin} \|u - \hat{a}\|^2 + \rho \xi^2$ subject to $L_f h_i + L_g h_i u \geq -\kappa h_i + \|L_g h_i\| \epsilon$ for every barrier i (hard constraints), plus $L_f V + L_g V u \leq -\gamma V + \|L_g V\| \epsilon + \xi$ when the fallback is active (soft, with slack ξ). Since $\dot{h} \geq L_f h + L_g h \cdot u - \|L_g h\| \epsilon$, a feasible QP guarantees $\dot{h} \geq -\kappa h$. So $h_i(x_0) \geq 0$ implies $h_i(x(t)) \geq 0$ for all t [15].

B4. Worked bond-dimension arithmetic (Gemma gate/up projection, 2048 $\rightarrow$ 16384, 33.55M parameters dense)

Target ratio	Bonds ($\chi_1, \chi_2, \chi_3, \chi_4$)	MPO params	Achieved	SVD rank at equal params
2×	32, 692, 692, 32	16.74M	2.00×	908
2.5×	32, 616, 616, 32	13.41M	2.50×	727
3×	32, 560, 560, 32	11.18M	3.00×	606
4×	32, 480, 480, 32	8.36M	4.01×	453

The table shows uniform allocation; step A3 redistributes bonds across layers. The outer bonds are capped at $32 = 8 \times 4$. Embedding: 257,152 tokens padded to 64^3 , width $2048 = 8 \cdot 16 \cdot 16$, three cores; $\chi = 226$ gives 52.6M parameters (10.0×). SigLIP widths (1152, 4304) are zero-padded to factorisable sizes, and padded entries are excluded from all counts. Layer-by-layer recount from the openpi configuration (parameters): Gemma MLP 1,811.9M; embedding 526.6M; SigLIP MLP 267.9M; Gemma attention 169.9M; SigLIP attention 143.5M; action expert 311.5M; other ≈ 8 M; total ≈ 3.24 B. The published figure is 3.3B [7]. At the §3 targets the model falls to ≈ 1.27 B, i.e. 2.55×

C. Validation Specifications

C1. Synthetic perturbation suite (§5.3)

Family	Parameter	Severity 1	Severity 2	Severity 3
Image noise	Gaussian σ (0–255 scale)	5	15	30
Motion blur	kernel length (px)	3	7	11
Illumination	brightness / contrast shift	$\pm 10\%$	$\pm 25\%$	$\pm 40\%$
Camera extrinsics	translation, rotation	2 cm, 3°	3.5 cm, 5.5°	5 cm, 8°
Proprioception	joint-position noise σ	0.005 rad	0.01 rad	0.02 rad
Actuation delay	control steps	1	2	3
Object dynamics	mass $\times$, friction $\times$	0.8–1.25, 0.8–1.2	0.6–1.6, 0.65–1.35	0.5–2.0, 0.5–1.5
External force	end-effector impulse, 0.2 s	5 N	10 N	20 N
Language	instruction paraphrase	synonym	reordered clause	novel phrasing

Safety evaluation uses LIBERO-Spatial and LIBERO-Long (the easiest and hardest suites): 10 tasks $\times$ 5 fixed initial states $\times$ 9 families $\times$ 3 severities = 2,700 episodes per arm per seed. If compute runs short, all families run at severity 2 and only camera extrinsics, actuation delay, external force and object dynamics run at all three severities.

C2. Metric definitions (fixed before any results)

- **Task success rate (SR):** fraction of episodes in which LIBERO's goal predicate holds within the suite's step limit. **Relative drop** = $(SR_{\text{ref}} - SR_{\text{arm}}) / SR_{\text{ref}}$, with absolute percentage points reported alongside. **Parameter ratio** = parameters in the reference $\div$ parameters in the deployed checkpoint, counting every tensor including embeddings and excluding padding.
- **Safe equilibrium:** at episode end, $\|q\| < 0.05$ rad/s held for ≥ 0.5 s, with $h_i(x) \geq 0$ for every barrier and no step in the episode where any $h_i < 0$. **Violation rate:** fraction of control steps with any $h_i < 0$. **Intervention rate:** fraction of steps with $\|u^* - \hat{a}\| > 10^{-3}$. **Coverage:** fraction of steps with $\|d\| \leq \epsilon_a$, reported per severity.
- **Latency:** p50 and p95 wall-clock time from receiving an observation to issuing the filtered command, batch size 1, measured over 1,000 cycles after 100 warm-up cycles. **Efficiency:** forward FLOPs per inference cycle (PyTorch profiler), and GPU-hours to reach reference SR on the held-out adaptation task.

- **Statistics:** mean $\pm$ SD over ≥ 3 seeds; 95% bootstrap CIs (10,000 resamples over task $\times$ seed); one-sided non-inferiority test with a 5% relative margin.

C3. Draft Resource Declaration (challenge \$6)

Item	Declared (planning estimate; finalised in the Phase 2 report)
GPU type and count	4–8 $\times$ A100/H100 80GB (cloud); 1 $\times$ RTX 4090; 1 $\times$ Jetson AGX Orin 64GB
Total GPU-hours	≈ 900 : compression and healing ≈ 530 ; LIBERO evaluation ≈ 100 ; safety suite ≈ 120 ; AD transfer ≈ 150
Simulation environments	LIBERO with robosuite/MuJoCo; Qiskit Aer (statevector and density-matrix) for the hardware pathway, ≤ 20 qubits. Exact versions pinned in the repository lock file
Quantum hardware / shots	None. Shot budget stated per pathway experiment
Estimated energy	≈ 430 kWh (400 W $\times$ 900 GPU-h $\times$ PUE 1.2); CO ₂ e from the cloud region's carbon intensity

D. References

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- Software: openpi and LeRobot (π_0), bitsandbytes, tntorch, quimb, OSQP, Qiskit Aer. All open-source; exact versions are pinned in the Phase 2 repository.